

Algebra II - Tutorial 3 - Date: 27/1/2025

1. Let $G = S_4$ and G' the symmetric group on the three elements of S_4 , $\Pi_1 = (12)(34)$, $\Pi_2 = (13)(24)$ and $\Pi_3 = (14)(23)$. Define $\phi : G \rightarrow G'$ as follows: $\phi(\sigma)$ is the permutation on $\{\Pi_1, \Pi_2, \Pi_3\}$ (Example 2.5.13) of Artin. Determine the kernel of ϕ . What are all the subgroups which contain K .
2. Let $G = \langle x \rangle$ be a cyclic group of order 12 and $G' = \langle y \rangle$ a group of order 6. Let $\phi : G \rightarrow G'$ be given by $\phi(x^i) = y^i$. Exhibit the correspondance referred to in the correspondance theorem explicitly.
3. Let G be a group of order p^2 . Show that it has atleast one subgroup of order p and if it has a unique subgroup of order p then it is a cyclic group.
4. Show that every subgroup of index 2 is a normal subgroup. Is a subgroup of index 3 a normal subgroup?